Knowledge Representation

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Part I

Foundation of Knowledge Representation

CHAPTER 1

Introduction to KR: Definition, Scope, Motivation (Use Cases), and History

1.1 | Introduction

What is Knowledge Representation?

Definition 1.1 Knowledge Representation
KR is a central problem in Artificial Intelligence that focuses on developing sufficiently precise notations for representing knowledge

The core problem with KR is that the real world with its complexity, ambiguity and dynamism can never be fully represented in a computer-based system. Nevertheless, intelligent systems must be able to function effectively using partial, incomplete and sometimes contradictory information. [4]

Why is Knowledge Representation needed?

The problem of data meaninglessness

To understand the need for KR, let's consider a practical scenario:

Example 1.1 A student database.

A hypothetical database contains the following entries:

Name	**Major**	**CGPA**	**Research**	...
Alice Smith	CS	3.8	Genomics	...
Bob John	Biology	3.8	Oncology	...
Carol Li	Zoology	3.7	Animal colonisation	...

Without knowledge representation, this data is disjointed. The entries exist in isolation across thousands of files. A database query system can only perform exact keyword matches:

- Query 1: "Show me all AI researchers" → No results (not explicitly mentioned in the data)
- Query 2: "Which students could work on this patent?" → No answer possible
- Query 3: "What are the emerging trends?" → No analysis possible

Data is meaningless without context and relationships. It does not allow for intelligent reasoning or deeper analysis.

What knowledge representation makes possible

With knowledge representation, we can structure the data and transfer isolated records into interconnected, queryable knowledge using:

- ontologys: Explicitly specify which concepts exist (student, researcher, field of research) and how they are categorised.
- Knowledge graphs: Link entities together: Alice → researcher → AI

This allows for semantic searches, automatic relationship discovery and reasoning across all connection.

Example 1.2

- Semantic search
  - "Find students researching machine learning?"
  - "Show me all AI researchers": The system can use transitive relationships to find relevant students.
  - "Which students are a good fit for this patent?": The system can match skills and areas of interest.
- Automatic relationship discovery
  - "What are emerging research trends?"
  - Recognising patterns across all data
  - Identifying emerging trends through interconnected information

Data	Information	Knowledge
Raw, unprocessed facts	Structured and contextualised data	Information combined with experience and reasoning rules
e.g. CGPA: 3.8	e.g. Alice has a CGPA of 3.8 in Computer Science	e.g. Alice could work on this AI patent because she has a strong foundation in computer science and is interested in genomics

Table 1.1: Data vs. Information vs. Knowledge

KR enables machines to make informed decisions with fewer rules.

KR Perspective

Traditional ML systems follow a black-box approach which means the decision-making process is not transparent and the reasoning behind predictions is not accessible.

- Generalisation: Requires continuous retraining with new data
- Explainability: Difficult or impossible to explain decisions
- Data flexibility: Highly dependent on large amounts of training data

Knowledge-Based AI utilises structured, explicit knowledge.

Definition 1.2 explicit knowledge
Knowledge that is formally encoded in symbols, rules or graphs and can be processed by machines

- Generalization: explicit knowledge transfers to new problems without retraining from scratch[3]
- Explainability: encoded knowledge enables transparent reasoning and human-understandable decisions[1]
- Data flexibility: Modifiable

Representation Schemes

Definition 1.3 representation scheme
A representation scheme is a formal notation used to define a knowledge base (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the 'language' in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.

Example 1.3 Pac-Man ghost AI has a knowledge-based approach: We code ghost behaviour as explicit rules.

1. IF player_nearby THEN chase
2. IF frightened THEN flee
3. ELSE patrol

Decisions are transparent, explainable, and modifiable without recomputing. (Ogland 2016)

Challenges in Knowledge Representation

Knowledge representation involves designing formal structures that allow computers to store, interpret, and reason about information and every design decision has advantages and caveats.

Choosing the representation is one of the design challenges. Semantic networks are intuitive and visually understandable but have limited inference. First-order logic on the other hand is expressive, but has computationally costly reasoning.

Desired properties can conflict. Systems should ideally be expressive enough to capture rich real-world knowledge while remaining efficient enough to reason in practical time. Tradeoffs are between expressiveness and tractability, and Human readability and machine efficiency.

Application-driven constraints arise from, different domains requiring different forms of representation. E.g. In natural language processing (NLP), systems must handle ambiguity and context. In medical diagnosis, representations must manage uncertainty, causal relationships, and strict correctness requirements. In question answering, systems need fast retrieval combined with reasoning over large amounts of structured and unstructured data

The core idea of Knowledge Representation is not just a data structure. It plays multiple roles simultaneously and those roles can conflict with each other:

- Surrogate for the world: Acts as a substitute for real world objects and relationships
- Ontological commitments: Defines what exists and how things are categorized
- Theory of reasoning: Provides foundation for logical inference and deduction
- Medium for efficient computation: Enables performant computational processing
- Medium of Human expression: Allows humans to express knowledge in understandable ways

1.2 | Taxonomy of Representation Schemes

Representational schemata can be divided into four basic categories:

- Declarative representations

  - Logical representation schemes
  - Semantic (network) schemes

- Procedural representations

- Hybrid approaches

- Frame-based representations

Declarative representations

Definition 1.4 declarative representation
Declarative representations describe facts about the world as logical statements

Declarative representations focus on truth and semantics and enable deductive reasoning. The schema is independent of how the knowledge is used.
There are two subtypes:

1. Logical representation schemes

Definition 1.5 logical representation scheme
Representation schema based on formal logic and represents knowledge through logical formulas. It is strongly based on semantics and inference

Example 1.4 First order Logic

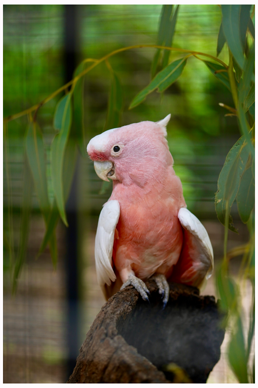

Figure 1.1: This is Tweety. What can we infer about Tweety with the information given below? (image from https://www.pexels.com/photo/pink-cockatoo-perched-in-natural-habitat-29732587/)

We know that all bird can fly if they are not flightless birds, here generalized as penguins. We know that Tweety is a bird and not a penguin.
More formal:

Birds can fly: $\forall x(Bird(x) \land \neg Penguin(x) \rightarrow CanFly(x))$
Facts: $Bird(Tweety)$ and $\neg Penguin(Tweety)$
Therefore we can infer:
$CanFly(Tweety)$

2. Semantic (network) schemes

Definition 1.6 semantic representation scheme
Graph-based representations in which nodes represent entities and edges represent relationships between them. It supports inheritance and classification

Example 1.5

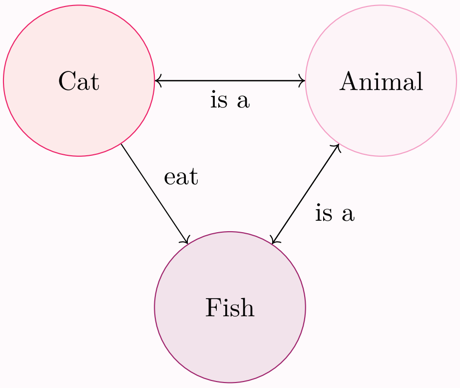

Figure 1.2: Semantic Network

Procedural representations

Definition 1.7 procedural representation
Representation schema that encodes knowledge as procedures or programmes that describe how to use that knowledge. It is efficient but hard to interpret

Example 1.6

Figure 1.3: This is Dr. Dokter. She wants to diagnose her patient. (image from https://www.pexels.com/photo/female-doctor-in-scrub-suit-7659874/)

The rules are:

IF (patient has fever) **AND** (patient has cough) **AND** (patient has difficulty breathing) **THEN diagnose:** possible pneumonia, **recommend** chest x-ray

This allows Dr. Dokter to properly diagnose her patients instead of furthering the antibiotic resistant strand of bacteria!

Hybrid approaches combine declarative and procedural aspects.

Frame-Based Representation

Definition 1.8 frame
Concept designed to provide a structured way to capture the essential aspects of a situation, facilitating easier retrieval and manipulation of information. They are akin to schemas or blueprints that organize knowledge into manageable chunks. Attributes of the represented concept are represented by slots which have facets and default values

Example 1.7 Back to Tweety from fig. 1.1.

Frame:	Bird	
	is_a:	Animal
	has_parts:	[wings, beak, feathers]
	can:	fly
	default_color:	pink
	locomotion:	flying

Frame:	Penguin	
	is_a:	Bird
	can:	swim
	cannot:	fly (overrides parent default)

1.3 | Distinguishing Features of KR Systems

Unlike databases (retrieval) or programs (execution), knowledge bases reuse the same facts in multiple ways:

Inference (rule-based reasoning):

Facts are used to derive new facts by applying rules.

Example 1.8 We remember Dr. Dokter from fig. 1.3.

$$\text{IF fever} \land \text{cough} \rightarrow \text{possible infection}$$

$\Rightarrow$ Diagnostic conclusion

It has multiple subtypes:

Definition 1.9 logical inference
Inference that uses formal deduction from known facts

Example 1.9

$$\text{IF fever} \rightarrow \text{infection}$$
$$\text{fever} \Rightarrow \text{infection}$$

Definition 1.10 specialized procedures
Fixed inference patterns, which are often implemented as routines

Example 1.10 Ward A $\rightarrow$ Ward B $\rightarrow$ ICU $\Rightarrow$ Shortest path

Definition 1.11 preferred inference
Reasoning based on typical assumptions, subject to certain reservations

Example 1.11

Birds fly
Tweety is a bird $\Rightarrow$ Tweety flies
(except penguins)

Definition 1.12 inductive inference
Generalising examples into patterns or rules

Example 1.12

Data $\Rightarrow$ Pattern
Symptoms $\Rightarrow$ Diagnosis model

Definition 1.13 abductive inference
Reasoning to arrive at the best explanation. From observations to hypothetical causes

Example 1.13

Observed: fever + cough
$\Rightarrow$ Best explanation: pneumonia

Access (Direct fact retrieval via queries):

Facts are retrieved directly, much like in a database.

Example 1.14 Query: "Which patients have fever?" $\Rightarrow$ Retrieve patient X

Matching (Similarity-based reasoning):

Facts are used to identify similar patterns and perform case-based reasoning.

Example 1.15

X $\sim$ Patient Y $\Rightarrow$ Case-based reasoning

Design implications

When designing a system, we must choose which inference mechanisms to support (e.g., shortest-path search vs. rule-based reasoning). Of course, there are trade-offs between expressiveness and efficiency. And different applications require different inference types, e.g. the Pac-Man game (example 1.3) needs heuristic inference for real-time decisions goals.

Observation 1.1 Knowledge bases cannot completely describe the world they model (except artificial microworlds)

Consequences exist for operations, reasoning and design methodology.

Implication 1.1 Regarding observation 1.1
There are different answers to existence queries depending on assumptions.
We can assume a closed world, where everything not explicitly stated in the knowledge base is assumed FALSE or an open world, where we simply cannot conclude anything if it is not explicitly stated in the knowledge base.

Example 1.16 To implication 1.1

1.4 | Applications of Knowledge Representation

Enterprise Knowledge Management

1.5 | Future Directions

Multi-hop reasoning Unified framework

knowledge Aggregation

def attention weight

1.6 | Course structure

foundation of KR

Knowledge graphs

KR in mL

grading:

50% of the exercise points (full portfolio) (Abgabe in 3er Gruppen(?)) present one exercise in tut present one paper in seminar 40–45 min (3er gruppen) final report 5-.10 pages (50% marks on final report)

Es wird ein Moodle geben

1.7 | Formalities

CHAPTER 2

Logical Foundations: First-Order Logic (FOL)

Knowledge representation requires a formal system that expresses knowledge precisely and derives consequences from that knowledge.[2]

The most relevant formal calculations are propositional Logic and first-order logic.

Definition 2.1 propositional Logic

todo

Example 2.1 "If a person is rich, they have a nice car"

That statement becomes

$$\begin{aligned} BobIsRich \Rightarrow \quad & BobHasNiceCar \\ JaneIsRich \Rightarrow \quad & JaneHasNiceCar \\ & \vdots \end{aligned}$$

Having to define every rule for every person is impractical and unintuitive though.

2.1 | Syntax

6 - Objects sind x,y, ...8 - not a well-formed formula weil es keinen Sinn macht

14 - unary predicate, welche gibt es noch??

Definition 2.2 formula

todo

Definition 2.3 atomic formula

todo

Definition 2.4 compound formula

todo

Definition 2.5 quantified formula

todo

Example 2.2 Building a Formula – Example from OLP

Definition 2.6 free

todo

Definition 2.7 bound
todo

Example 2.3 Von f 18

Definition 2.8 sentence
todo

Example 2.4 Von f 19

Definition 2.9 substitution
todo

Example 2.5 Von f 20.1

Definition 2.10 captured
todo

Example 2.6 Von f 21

2.2 | Semantics

Definition 2.11 language structure
todo

Definition 2.12 domain
todo

Definition 2.13 valid
todo

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todo

Definition 2.15 unsatisfiable
todo

2.3 | Inference

Definition 2.16 inference
todo

Definition 2.17 Modus Ponens
todo

Definition 2.18 Modus Tollens
todo

Definition 2.19 Universal Instantiation
todo

Definition 2.20 Existential Instantiation
todo

Definition 2.21 Prenex Normal Form
todo

F 38 in step 1: Rule 1: $A \to B = \neg A \cup B$
Rule 2: $\neg \forall x(P(x)) \equiv \ldots$ Ist es hier neg oder sim ??

Definition 2.22 clause
todo

Definition 2.23 literal
todo

Definition 2.24 Conjunctive Normal Form
todo

Definition 2.25 Resolution-based prover
todo

Definition 2.26 resolution
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Definition 2.28 Sequent Calculus
todo

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todo

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Part II

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CHAPTER 5

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CHAPTER 6

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CHAPTER 7

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Part III

Knowledge Representation in Machine Learning

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